

Ex. No. : 06

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A Python Program To Do Face Recognition Using SVM Classifier

Aim:

To implement **face recognition using Support Vector Machine (SVM)**.

Requirements:

Jupyter Notebook with Python, fetch_olivetti_faces dataset, and required Python libraries.

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_lfw_people
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import classification_report

lfw = fetch_lfw_people(min_faces_per_person=50, resize=0.4)

X = lfw.data
y = lfw.target
images = lfw.images
names = lfw.target_names

print("Classes:", names)

X_train, X_test, y_train, y_test, img_train, img_test = train_test_split(
    X, y, images, test_size=0.25, random_state=42
)

model = SVC(kernel='rbf', class_weight='balanced')
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred, target_names=names))
```

```
fig, axes = plt.subplots(3, 5)
```

```
for i, ax in enumerate(axes.flat):
    ax.imshow(img_test[i])
    ax.set_title(names[y_pred[i]])
    ax.axis('off')
```

```
plt.suptitle("Face Recognition using SVM")
plt.show()
```

CLASSIFICATION REPORT :

```
Classes: ['Ariel Sharon' 'Colin Powell' 'Donald Rumsfeld' 'George W Bush'
'Gerhard Schroeder' 'Hugo Chavez' 'Jacques Chirac' 'Jean Chretien'
'John Ashcroft' 'Junichiro Koizumi' 'Serena Williams' 'Tony Blair']
```

```
Classification Report:
```

	precision	recall	f1-score	support
Ariel Sharon	0.65	0.81	0.72	16
Colin Powell	0.78	0.85	0.81	66
Donald Rumsfeld	0.63	0.70	0.67	27
George W Bush	0.92	0.73	0.81	140
Gerhard Schroeder	0.64	0.67	0.65	24
Hugo Chavez	0.76	0.76	0.76	17
Jacques Chirac	0.57	0.80	0.67	10
Jean Chretien	0.92	0.79	0.85	14
John Ashcroft	0.67	0.88	0.76	16
Junichiro Koizumi	0.75	0.82	0.78	11
Serena Williams	0.80	0.89	0.84	9
Tony Blair	0.65	0.75	0.70	40
accuracy			0.77	390
macro avg	0.73	0.79	0.75	390
weighted avg	0.79	0.77	0.77	390

OUTPUT:



Result:

Thus, the Multi Layer Perceptron with Backpropagation was successfully implemented and evaluated using the dataset.